

BOSTOVO CHEREPOVETS, Russian Federation

WMO: 271130

Lat: **59.2826N**Lon: **38.0256E**Elev: **385**StdP: **14.49**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-14.9	-26.0	1.3	-20.5	-19.6	1.9	-14.7	20.2	24.4	18.4	21.0	1.8	50	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	82.8	68.0	79.3	66.3	75.9	64.5	70.1	79.5	68.2	76.6	66.2	73.6	7.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	100.0	74.9	64.9	93.5	72.4	63.2	87.9	70.4	34.3	79.9	32.7	76.9	31.1	73.5	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	19.3	16.2	13.7	DB	-28.2	87.2	6.6	3.4	-33.0	89.6	-36.8	91.6	-40.5	93.5	-45.3	96.0
(5)				WB	-28.5	73.0	6.5	2.7	-33.1	74.9	-37.0	76.5	-40.6	78.0	-45.4	80.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	14.1	15.2	24.6	38.4	51.1	58.5	63.8	59.2	50.0	38.8	27.6	19.7
	DBStd	19.84	14.42	14.23	9.70	8.06	8.48	7.05	6.39	6.16	6.75	7.50	10.53	13.65
	HDD50	5383	1114	974	788	357	91	9	1	4	81	351	673	940
	HDD65	9766	1579	1394	1253	797	437	218	100	200	449	811	1123	1405
	CDD50	1201	0	0	0	10	124	264	429	288	82	4	0	0
	CDD65	112	0	0	0	0	5	24	63	20	0	0	0	0
	CDH74	1167	0	0	0	2	86	280	586	209	4	0	0	0
	CDH80	270	0	0	0	0	13	61	151	45	0	0	0	0
Wind	WSAvg	5.9	6.0	6.3	6.3	5.9	6.2	5.5	4.9	4.9	5.1	6.3	6.5	6.6
Precipitation	PrecAvg	35.0	3.5	2.6	2.4	1.7	2.3	3.0	3.2	3.1	2.7	3.2	3.7	3.7
	PrecMax	45.3	5.4	4.7	4.4	3.2	4.1	5.2	5.9	6.5	4.9	6.0	6.6	5.7
	PrecMin	27.0	1.4	0.7	0.4	0.7	0.7	0.7	0.3	0.8	1.5	1.1	0.5	1.6
	PrecStd	4.3	1.0	1.0	0.9	0.7	0.7	1.0	1.6	1.4	0.9	1.3	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	38.2	49.2	71.0	82.1	85.9	89.3	85.1	73.9	62.1	47.7	43.1
		MCWB	35.2	35.5	40.6	54.8	64.3	68.5	72.7	69.8	63.1	54.6	45.7	42.1
	2%	DB	35.0	35.9	42.8	64.0	75.5	81.3	83.9	79.1	68.5	54.6	44.1	39.3
		MCWB	33.7	34.0	36.9	50.6	61.1	66.8	69.2	66.6	60.6	50.8	42.7	38.1
	5%	DB	33.7	34.3	39.3	58.6	71.1	77.1	80.3	75.3	64.6	51.9	41.5	36.2
		MCWB	32.8	32.7	35.1	48.0	57.0	64.2	67.6	64.6	58.2	49.4	40.3	35.0
	10%	DB	32.3	32.9	36.9	52.8	67.0	72.8	76.6	71.5	61.1	49.5	39.4	34.1
		MCWB	31.4	31.6	34.0	44.5	54.9	62.0	65.8	62.8	56.1	47.5	38.4	33.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.7	36.2	41.7	55.1	66.7	71.2	73.4	71.9	64.5	55.5	46.3	42.1
		MCDB	36.1	37.4	47.3	69.2	76.4	81.0	84.9	82.3	71.3	60.5	47.0	42.7
	2%	WB	33.9	34.4	38.0	51.7	63.0	68.5	71.0	68.4	61.8	51.8	42.9	38.2
		MCDB	34.5	35.5	41.1	61.2	73.2	78.4	80.7	76.3	67.1	53.5	43.8	39.0
	5%	WB	32.8	33.0	35.9	48.6	59.6	65.9	69.0	66.1	59.2	49.6	40.4	35.3
		MCDB	33.6	34.1	38.2	56.7	66.9	74.4	77.8	73.0	63.4	51.4	41.3	36.1
	10%	WB	31.4	31.7	34.3	45.4	56.5	63.4	67.2	64.1	56.8	47.6	38.4	33.2
		MCDB	32.3	32.9	36.3	52.3	64.4	70.7	74.7	70.1	60.5	49.3	39.3	33.8
Mean Daily Temperature Range	5% DB	MDBR	11.7	13.8	16.1	17.9	19.9	19.0	18.8	18.6	15.4	9.8	7.5	9.3
		MCDBR	7.2	8.9	17.6	28.2	27.0	25.4	24.5	24.7	21.7	14.4	8.2	6.2
		MCWBR	7.5	8.2	12.7	16.2	13.1	12.0	11.3	13.1	13.7	10.6	7.6	6.2
	5% WB	MCDBR	7.4	8.4	12.9	25.5	22.6	21.9	21.9	21.9	19.3	12.4	7.7	6.0
		MCWBR	7.7	8.1	9.9	15.4	12.0	11.3	10.8	12.3	12.9	9.7	7.5	6.1
Clear-Sky Solar Irradiance	taub	0.269	0.313	0.334	0.379	0.365	0.361	0.395	0.392	0.356	0.334	0.296	0.253	
	taud	2.062	2.060	2.108	2.208	2.408	2.439	2.363	2.351	2.431	2.371	2.167	2.028	
	Ebn at Noon	160	212	249	259	274	277	263	254	246	211	150	121	
	Edh at Noon	16	27	35	38	33	33	35	33	26	20	14	11	
All-Sky Solar Radiation	RadAvg	115	345	810	1234	1635	1727	1685	1296	756	325	105	62	
	RadStd	17	73	110	147	106	108	209	87	82	56	19	10	